Toric Varieties

Labix

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Abstract

Contents

1 Monoids

1.1 Basic Properties of Monoids

Definition 1.1.1 (Monoids)
A monoid is a pair $(M, *)$ where M is a set and $* : M \times M \to M$ is a binary operation on M satisfying the following axioms:
- Associativity: $(a * b) * c = a * (b * c)$ for all $a, b, c \in M$.
- Identity: There exists $1_M \in M$ such that $1_M * a = a = a * 1_M$ for all $a \in M$.

Definition 1.1.2 (Monoid Homomorphisms)
Let M, N be monoids. Let $\phi : M \to N$ be a function. We say that ϕ is a monoid homomorphism if the following are true.
- $\phi(m_1 m_2) = \phi(m_1)\phi(m_2)$ for all $m_1, m_2 \in M$.
- $\phi(1_M) = 1_N$.

Definition 1.1.3 (Commutative Monoids)
Let M be a monoid. We say that M is commutative if

$$a * b = b * a$$

for all $a, b \in M$.

Definition 1.1.4 (Cancellative Monoids)
Let M be a monoid. We say that M is cancellative if the following are true.
- $ab = ac$ implies $b = c$ for all $a, b, c \in M$.
- $ba = ca$ implies $b = c$ for all $a, b, c \in M$.

Definition 1.1.5 (Finitely Generated Monoids)
Let M be a monoid. We say that M is finitely generated if there exists $m_1, \dots, m_n \in M$ such that

$$m = m_1^{a_1} \cdots m_n^{a_n}$$

for all $m \in M$ and some $a_1, \dots, a_n \in \mathbb{N}$.

1.2 The Grothendieck Completion of a Group

Definition 1.2.1 (Grothendieck Completion)
Let M be a commutative monoid. Define the Grothendieck completion of M to be the set

$$G(M) = \frac{M \times M}{\sim}$$

where $(m_1, m_2) \sim (n_1, n_2)$ if $m_1 + n_2 + k = m_2 + n_1 + k$ together with the binary operation $* : G(M) \times G(M) \to G(M)$ defined by

$$(m_1, m_2) * (n_1, n_2) = (m_1 + n_1, m_2 + n_2)$$

Lemma 1.2.2
Let M be a commutative monoid. Then $G(M)$ is an abelian group.

Definition 1.2.3 (The Canonical Homomorphism)
Let M be a commutative monoid. Define the canonical homomorphism to the Grothendieck completion to be the function $i : M \to G(M)$ given by $i(m) = (m, 0)$.

Proposition 1.2.4 (Universal Property of the Grothendieck Completion)
Let M be a commutative monoid. Then $G(M)$ and the canonical map $i : M \to G(M)$ satisfies the following universal property: For any abelian group G and any monoid homomorphism $\phi : M \to G$, there exists a unique group homomorphism $\psi : G(M) \to G$ such that the following diagram commutes:

$$\begin{tikzcd} M \arrow[r, "i"] \arrow[dr, "\phi"'] & G(M) \arrow[d, dashed, "\exists!\psi"] \\ & G \end{tikzcd}$$

Moreover, $G(M)$ is the unique (up to unique isomorphism) abelian group satisfying this property.

Lemma 1.2.5
Let M be a cancellative commutative monoid. Then the canonical morphism $i : M \to G(M)$ is injective.

1.3 The Monoid Ring

Definition 1.3.1 (The Monoid Ring)
Let S be a monoid. Let R be a ring. Define the semi-group ring of S over R to be the left R-module

$$R[S] = \bigoplus_{s\in S} R \cdot t^s$$

together with multiplication defined on basis by $t^{s_1} \cdot t^{s_2} = t^{s_1+s_2}$

Lemma 1.3.2
Let S be a monoid. Let R be a ring. Then the following are true.

- $R[S]$ is a ring.
- If S is commutative, then $R[S]$ is a commutative ring.
- If R and S are commutative, then $R[S]$ is an R-algebra.

Definition 1.3.3 (Monomial Notation)
Let R be a commutative ring. Let $u = (u_1, \dots, u_n) \in \mathbb{N}^n$. Define

$$x^u = x_1^{u_1} \cdots x_n^{u_n} \in R[x_1, \dots, x_n]$$

Definition 1.3.4 (Lattice Ideal)
Let $L \subseteq \mathbb{Z}^n$ be a lattice. Let R be a commutative ring. Define the lattice ideal of L in $R[x_1, \dots, x_n]$ to be

$$I_L = \langle x^u - x^v \mid u, v \in \mathbb{N}^n \text{ and } u - v \in L\rangle$$

Proposition 1.3.5
Let k be a field. Let S be a finitely generated commutative monoid. Let $s_1, \dots, s_n$ be a set of generators of S. Let $\phi : \mathbb{Z}^n \to G(S)$ be the group homomorphism defined by $e_i \mapsto s_i$. Then there is a k-algebra isomorphism

$$k[S] \cong \frac{k[x_1, \dots, x_n]}{I_{\ker(\phi)}}$$

Proposition 1.3.6
Let k be a field. Let S be a finitely generated commutative monoid. Let $s_1, \dots, s_n$ be a set of generators of S. Then we have

$$\dim(k[S]) = n - \text{rank}(\ker(\phi))$$

Definition 1.3.7 (Affine Semi-group)
Let S be a monoid. We say that S is an affine semi-group if there exists an injective monoid homomorphism $S \to \mathbb{N}^d$ for some $d \in \mathbb{N}$.

Proposition 1.3.8
Let k be a field. Let S be a finitely generated commutative monoid. Let $s_1, \dots, s_n$ be a set of generators of S. Let $\phi : \mathbb{Z}^n \to G(S)$ be the group homomorphism defined by $e_i \mapsto s_i$. Then the following are equivalent.

- $I_{\ker(\phi)}$ is prime.
- $k[S]$ is an integral domain.
- $G(S)$ is a free abelian group.
- S is an affine semi-group.

1.4 Affine Semigroup Associated to a Rational Polyhedral Cone

Lemma 1.4.1 (Gordon's Lemma)
Let $\sigma \subseteq \mathbb{R}^n$ be a rational polyhedral cone. Let $A \subseteq \mathbb{Z}^n$ be a subgroup. Then $\sigma \cap A$ is an affine semi-group.

Conversely, given an affine semigroup $S \subseteq \mathbb{N}^n$, we may associate to it a cone

$$\mathbb{R}_{\geq 0} S = \{rs \mid r \in \mathbb{R}_{\geq 0} \text{ and } s \in S\}$$

So given an affine semigroup, we may obtain a new affine semigroup by passing to cones then passing back. This is called the saturation of the affine semigroup.

Definition 1.4.2 (Saturation of an Affine Semigroup)
Let S be an affine semigroup. Define the saturation of S to be

$$S_{\text{sat}} = (\mathbb{R}_{\geq 0} S) \cap G(S)$$

Proposition 1.4.3
Let S be an affine semigroup. Then we have

$$\overline{k[S]} = k[S_{\text{sat}}]$$

Definition 1.4.4 (Saturated Affine Semigroup)
Let S be an affine semigroup. We say that S is saturated if $S = S_{\text{sat}}$.

Lemma 1.4.5
Let $\sigma \subseteq \mathbb{R}^n$ be a strongly convex rational polyhedral cone. Then $\sigma \cap S$ is saturated.

Lemma 1.4.6
Let $\sigma_1, \sigma_2 \subseteq \mathbb{R}^n$ be cones such that $\sigma_1 \cap \sigma_2$ is a face of both cones. Let $A \subseteq \mathbb{Z}^n$ be a subgroup. Then we have

$$(\sigma_1 \cap \sigma_2) \cap A = (\sigma_1 \cap A) + (\sigma_2 \cap A)$$

2 Introduction to Toric Varieties

2.1 The Standard n-Torus

Definition 2.1.1 (The Standard n-Torus)
Let k be a field. Let $n \in \mathbb{N}^+$. Define the standard n-torus over k to be the affine group variety

$$\mathbb{G}_m^n(k) = \text{Spec}(k[x, x^{-1}])^n$$

Definition 2.1.2 (Tori)
Let k be a field. Let T be an affine variety over k. We say that T is a torus if there exists a field extension K/k such that $T \cong \mathbb{G}_m^n(K)$.

Definition 2.1.3 (Split Tori)
Let k be a field. Let T be a torus over the field extension K/k. We say that T is split if $K = k$.

If K is separably closed or algebraically closed, then all tori are split.

Lemma 2.1.4
Let k be a field. Let T be a torus over k. Then the following are true.
- Let T' be a torus over k. Let $\phi : T \to T'$ be a morphism of group schemes. Then $\text{im}(\phi)$ is closed torus in T'.
- Let $H \leq T$ be a subgroup variety of T. If H is irreducible, then H is a torus.
- Let $H \leq T$ be a subgroup of T. Then T/H is a torus.

Recall that the functor $D : \textbf{FGAb} \to \textbf{Diag}$ sending a finitely generated abelian group G to $D(G) = \text{Spec}(k[G])$ is an equivalence of categories. Evidently split tori corresponds to the free abelian groups of finite rank under the equivalence.

The following are some more properties of a split torus T:

- T is diagonalizable (T is a torus if and only if T is smooth connected diagonalizable)
- Every representation of G is diagonalizable.
- If V is a representation of T, then we can decompose V into its eigenspaces:
$$V = \bigoplus_{\chi \in \chi(T)} V_\chi$$

Let $(m_1, \dots, m_n) \in \mathbb{Z}^n$, we have a character $\chi : T \to \mathbb{C}^\times$ defined by $(t_1, \dots, t_n) \mapsto t_1 m_1 \cdots t_n^{m_n}$. All characters of T are of this form. In other words, we have $M = \text{Hom}(T, \mathbb{C}^\times) \cong \mathbb{Z}^n$.

Definition 2.1.5 (One-parameter Subgroups)
Let k be a field. Let T be a torus over k. A one-parameter subgroup of T is a homomorphism of group schemes $\mathbb{G}_m \to T$.

Definition 2.1.6 (The Group of One-Parameter Subgroups)
Let k be a field. Let T be a torus over k. Define the group of one parameter subgroups of T to be the set

$$\lambda(T) = \{\lambda : \mathbb{G}_m \to T \mid \lambda \text{ is a one-parameter subgroup } \}$$

together with pointwise multiplication.

Lemma 2.1.7
Let k be a field. Let T be a torus over k. Let $f : T \to \mathbb{G}_m^n$ be an isomorphism. Then f induces isomorphisms

$$\chi(T) \cong \mathbb{Z}^n \quad \text{and} \quad \lambda(T) \cong \mathbb{Z}^n$$

Lemma 2.1.8
Let k be a field. Let T be a split torus over k. Then the map

$$\langle -,-\rangle : \chi(T)\times\lambda(T)\to\chi(\mathbb{G}_m)\cong\mathbb{Z}$$

given by $\langle\chi,\lambda\rangle\mapsto\chi\circ\lambda$ is a perfect pairing.

2.2 General Toric Varieties

Definition 2.2.1 (Toric Varieties)
Let X be an irreducible variety over $\mathbb{C}$. We say that X is a toric variety if the following are true.

- X contains a torus T_X as a Zariski open subset.
- The action of T_X on itself extends to X.

The idea is that toric varieties consists of the underlying points together with limit points. Recall that there is a vector space decomposition

$$\mathbb{C}[T]\cong\bigoplus_{\chi\in\chi(T)}\mathbb{C}\cdot\chi$$

This means that the character functions span coordinate ring of the torus as a vector space.

The one parameter subgroups $\lambda\in\lambda(T)$ traces out the curves on the torus. To add in limit points to the torus T is the same as adding in the limit $\lim_{t\to 0}\lambda(t)$ since λ is defined on $\mathbb{C}\setminus\{0\}$.

For example, suppose we wish to add in the point $(1,0)\in\mathbb{C}^2$ to the torus $T=(\mathbb{C}^\times)^2$. When we add int limit points, we expect certain characters on the torus to no longer be elements of the coordinate ring of the toric variety. Let $x^ay^b\in\mathbb{C}[T]$ be a general character of a torus. Then it is well defined on the point $(1,0)$ if and only if $b>0$. This adds a linear inequality to the lattice $\chi(T)$. On the other hand, we may think of $(1,0)$ as the limit point of the one parameter subgroup $\lambda(t)=(t^1,t^0)$. However we also need to consider the equivariant action of the torus, so adding just the point $(1,0)$ is not enough.

Example 2.2.2
The following are true.

- $\mathbb{C}^\times$ is a toric variety.
- $\mathbb{C}^n$ is a toric variety.
- $\mathbb{P}^n$ is a toric variety.
- If X,Y are toric varieties, then $X\times Y$ is a toric variety.
- The total space of the line bundle over $\mathbb{P}^n$ is a toric variety.

The equivariant structure of the torus encodes a lot of the algebraic geometry into combinatorial data or more managable structure. A lot of common varieties are toric, or are subvarieties of complete intersections of toric varieties.

There are two ways to construct normal toric varieties:

(1): Make an open cover of it via affine schemes that each corresponds to some polyhedral cone, then glue the open cover from combinatorial data.

(2): Encode a projective variety as the proj construction given by a polytope.

2.3 Constructing Affine Toric Varieties

Definition 2.3.1 (Affine Toric Varieties Associated to Characters)
Let T be a torus over $\mathbb{C}$. Let $\mathcal{A}=\{\chi_1,\dots,\chi_s\}\subseteq\chi(T)$ be characters of T. Define the affine toric variety associated to $\mathcal{A}$ to be

$$Y_{\mathcal{A}}=\overline{\operatorname{im}(\Phi_{\mathcal{A}})}$$

where $\Phi_\mathcal{A} : T \to \mathbb{A}^s_\mathbb{C}$ is the map defined by $\Phi_\mathcal{A}(t) = (\chi_1(t), \dots, \chi_s(t))$

Proposition 2.3.2
Let T be a torus over $\mathbb{C}$. Let $\mathcal{A} = \{\chi_1, \dots, \chi_s\} \subseteq \chi(T)$ be characters of T. Then the following are true.

- $Y_\mathcal{A}$ is an affine toric variety.
- $\chi(T_{Y_\mathcal{A}}) = \langle \mathcal{A} \rangle$.
- $\dim(Y_\mathcal{A}) = \text{rank}(\langle \mathcal{A} \rangle)$.
- The ideal of $Y_\mathcal{A}$ is given by
$$\mathbb{I}(Y_\mathcal{A}) = I_{\ker(\phi)} = \langle x^\alpha - x^\beta \mid \alpha, \beta \in \mathbb{N}^s, \alpha - \beta \in \ker(\phi) \rangle$$
where $\phi : \mathbb{Z}^s \to \chi(T)$ is the group homomorphism defined by $e_i \mapsto \chi_i$.

Proof
Notice that the image of $\Phi_\mathcal{A}$ lands in $\mathbb{G}^s_m$. Hence $\Phi_\mathcal{A}$ is a map of torii. By lmm we know that $\text{im}(\Phi_\mathcal{A})$ is a closed torus in $\mathbb{G}^s_m$. By the relative closure formula, we have that
$$Y_\mathcal{A} \cap \mathbb{G}^s_m = \text{im}(\Phi_\mathcal{A})$$
Then by definition of subspace topology on $Y_\mathcal{A}$, we conclude that $\text{im}(\Phi_\mathcal{A})$ is an open subset of $Y_\mathcal{A}$. This shows that $Y_\mathcal{A}$ contains a torus as an open subset. Also, since $\text{im}(\Phi_\mathcal{A})$ is irreducible and is an open subset of $Y_\mathcal{A}$, we conclude that $Y_\mathcal{A}$ is irreducible.

Since T acts on itself for being a torus, the map $\Phi_\mathcal{A}$ gives the action of $\text{im}(\Phi_\mathcal{A})$ on $\mathbb{G}^s_m$ that sends varieties to varieties. Let $t \in \text{im}(\Phi_\mathcal{A})$. Since $\text{im}(\Phi_\mathcal{A})$ is a torus, we have that
$$\text{im}(\Phi_\mathcal{A}) = t \cdot \text{im}(\Phi_\mathcal{A}) \subseteq t \cdot Y_\mathcal{A}$$
By definition of closure, we must have $Y_\mathcal{A} \subseteq t \cdot Y_\mathcal{A}$. Replacing t with t^{-1} show that $t \cdot Y_\mathcal{A} \subseteq Y_\mathcal{A}$. Hence the action of $\text{im}(\Phi_\mathcal{A})$ gives an action on $Y_\mathcal{A}$. Thus $Y_\mathcal{A}$ is an affine toric variety. ∎

Example 2.3.3 (The Rational Normal Cone)
Let $d \in \mathbb{N}^+$. Let $\phi : \mathbb{C}^2 \to \mathbb{C}^{d+1}$ be the morphism defined by
$$\phi(s, t) = (s^d, s^{d-1}t, \dots, st^{d-1}, t^d)$$
Then the following are true.

- $\text{im}(\phi)$ is an affine toric variety.
- $\mathbb{I}(\text{im}(\phi)) = (x_i x_{j+1} - x_{i+1} x_j \mid 1 \leq i < j \leq d)$.

Proof
Let $I = (x_i x_{j+1} - x_{i+1} x_j \mid 1 \leq i < j \leq d) \subseteq \mathbb{C}[x_1, \dots, x_{d+1}]$. Notice that by projecting to projective space, we have that ϕ is the veronese map. It follows from the same technique that $\text{im}(\phi) = \mathbb{V}(I)$. By the above, we know that $\text{im}(\phi)$ is an affine toric variety.

Now we have $\mathbb{I}(\text{im}(\phi)) = \mathbb{I}(\mathbb{V}(I)) = I$ because toric varieties are irreducible. ∎

Definition 2.3.4 (Toric Ideals)
Let $L \subseteq \mathbb{Z}^n$ be a lattice. Let R be a commutative ring. Let $I_L \subseteq R[x_1, \dots, x_n]$ be the lattice ideal of L. We say that I_L is a toric ideal if I_L is a prime ideal.

Lemma 2.3.5
Let $I \subseteq \mathbb{C}[x_1, \dots, x_n]$ be an ideal. Then I is toric if and only if it is prime and is generated by binomials of the form $x_{i_1}^{\alpha_1} \cdots x_{i_n}^{\alpha_n} - x_{j_1}^{\beta_1} \cdots x_{j_m}^{\beta_m}$.

Proof
If I is a toric ideal then it is clearly prime and generated by binomials. So suppose that I is prime and is generated by binomials of the above form. Notice that $\mathbb{V}(I) \cap (\mathbb{C}^\times)^n$ is non-empty since it contains $(1, \dots, 1)$. We check that it is a subgroup of $(\mathbb{C}^\times)^n$. Let $u, v \in \mathbb{V}(I) \cap (\mathbb{C}^\times)^n$. Then we know that $u^\alpha - u^\beta = 0 = v^\alpha - v^\beta$. Then we have

$$(uv)^\alpha - (uv)^\beta = u^\alpha v^\alpha - u^\beta v^\alpha + u^\beta v^\alpha - u^\beta v^\beta = (u^\alpha - u^\beta)v^\alpha + u^\beta(v^\alpha - v^\beta) = 0$$

Hence products are closed. Also, we have Hence inverses are closed. Thus $\mathbb{V}(I) \cap (\mathbb{C}^\times)^n$ is a subgroup of $(\mathbb{C}^\times)^n$. By lmm, we conclude that $\mathbb{V}(I) \cap (\mathbb{C}^\times)^n$ is a torus. For each $1 \leq i \leq n$, projecting to the ith coordinate gives a character $\mathbb{V}(I) \cap (\mathbb{C}^\times)^n \to (\mathbb{C}^\times)^n \to \mathbb{C}^\times$. Let $\mathcal{A}$ be the set of these characers. Then by construction we have $\mathbb{V}(I) = Y_\mathcal{A}$. By the Nullstellensatz, we have $I = \mathbb{I}(\mathbb{V}(I)) = \mathbb{I}(Y_\mathcal{A})$. Hence by the above we conclude that I is toric. ■

Proposition 2.3.6
Let S be an affine semigroup. Then the following are true.
- Spec$(\mathbb{C}[S])$ is an affine toric variety.
- The character group of the torus in Spec$(\mathbb{C}[S])$ is given by $\langle t^s \mid s \in S \rangle$.

Proposition 2.3.7
Let X be an irreducible affine variety over $\mathbb{C}$. Then the following are equivalent.
- X is a affine toric variety.
- There exists a torus T and a finite subset $\mathcal{A} \subseteq \chi(T)$ such that $X = Y_\mathcal{A}$.
- There exists a toric ideal I of $\mathbb{C}[x_1, \dots, x_n]$ such that $X \cong$ Spec $\left(\frac{\mathbb{C}[x_1,\dots,x_n]}{I}\right)$.
- There exists an affine semigroup S such that $X \cong$ Spec$(\mathbb{C}[S])$.

Let S be an affine semigroup. Recall that the saturation of S is the affine semigroup

$$S_{\text{sat}} = \mathbb{R}_{\geq 0} S \cap G(S)$$

where $G(S)$ is the Grothendieck completion of S.

Lemma 2.3.8
Let $X =$ Spec$(\mathbb{C}[S])$ be an affine toric variety over $\mathbb{C}$ where S is an affine semigroup. Then the normalization of X is given by

$$\widetilde{X} = \text{Spec}(\mathbb{C}[S_{\text{sat}}])$$

together with the normalization map given by the inclusion $\mathbb{C}[S] \hookrightarrow \mathbb{C}[S_{\text{sat}}]$.

Proposition 2.3.9
Let k be an algebraically closed field. Let S be an affine semi-group with generators given by $m_1, \dots, m_s$. Then there is a bijection

$$\{p \in \text{Spec}(k[S]) \mid p \text{ is a closed point }\} \overset{1:1}{\longleftrightarrow} \{\phi : S \to (\mathbb{C}, \cdot) \mid \phi \text{ is a monoid homomorphism }\}$$

given as follows.
- Each closed point $p \in$ Spec$(k[S])$ corresponds to the monoid homomorphism $\gamma_p : S \to \mathbb{C}$ defined by
$$\gamma_p\left(\sum_{i=1}^s \lambda_i m_i\right) = \sum_{i=1}^s \lambda_i x_i(p)$$
where $x_i \in \mathbb{C}[S]$ is the ith coordinate function.
- Each monoid homomorphism $\gamma : S \to (\mathbb{C}, \cdot)$ corresponds to the point $(\gamma(m_1), \dots, \gamma(m_s))$.

2.4 Cones and Affine Normal Toric Varieties

Definition 2.4.1 (Affine Semigroup Associated to a Cone)
Let $\sigma \subseteq \mathbb{R}^n$ be a rational polyhedral cone. Define the associated affine semigroup to be

$$S_\sigma = \sigma^\vee \cap (\mathbb{R}^n)^*$$

Example 2.4.2
There is an isomorphism
$$\text{Spec}(\mathbb{C}[\{0\}^\vee \cap \mathbb{Z}^n]) \cong \text{Spec}(\mathbb{G}_m^n)$$

Proof
The dual cone of $\{0\}$ is $\mathbb{R}^n$ and intersecting with $\mathbb{Z}^n$ just gives itself. So $\mathbb{C}[\mathbb{Z}^n] = \mathbb{C}[x_1, \dots, x_n, x_1^{-1}, \dots, x_n^{-1}]$ is just the coordinate ring of the standard n-torus. ■

Example 2.4.3
Let $e_1, e_2, e_3 \in \mathbb{R}^3$ be the standard basis vectors. Let $\sigma = \text{Cone}(e_1, e_2, e_1 + e_3, e_2 + e_3) \subseteq \mathbb{R}^3$. Then we have
$$\mathbb{C}[S_\sigma] = \frac{\mathbb{C}[x, y, z, w]}{(xy - zw)}$$

Proof
We compute the dual cone to be

$$\sigma^\vee = H^+_{(1,0,0)} \cap H^+_{(0,1,0)} \cap H^+_{(1,0,1)} \cap H^+_{(0,1,1)}$$

A generating set of S_σ as a monoid is given by $\{e_1, e_2, e_3, e_1 + e_2 - e_3\}$. The kernel of the map $\mathbb{Z}^4 \to \mathbb{Z}^3$ defined by $e_i \mapsto e_i$ for $1 \leq i \leq 3$ and $e_4 \mapsto (1, 1, -1)$ is given by $\mathbb{Z}\langle e_1 + e_2 - e_3 - e_4 \rangle$. Thus we are done. ■

Example 2.4.4
Let $e_1, e_2, e_3 \in \mathbb{R}^3$ be the standard basis vectors. Let $\sigma = \text{Cone}(e_2, e_3, 2e_1 + e_2 + e_3) \subseteq \mathbb{R}^3$. Then we have
$$\mathbb{C}[S_\sigma] \cong \frac{\mathbb{C}[x_1, \dots, x_6]}{(x_1x_4 - x_2x_3, x_1x_5 - x_2^2, x_1x_6 - x_3^2)}$$

Proof
We compute the dual cone to be

$$\sigma^\vee = H^+_{(2,1,1)} \cap H^+_{(0,1,0)} \cap H^+_{(0,0,1)}$$

A generating set of S_σ as a monoid is then given by $\{(1, 0, 0), (0, 1, 0), (0, 0, 1), (-1, 1, 1), (-1, 2, 0), (-1, 0, 2)\}$. The kernel of the map $\mathbb{Z}^6 \to \mathbb{Z}^3$ defined by $e_i \mapsto e_i$ for $1 \leq i \leq 3$ and $e_4 \mapsto (-1, 1, 1), e_5 \mapsto (-1, 2, 0), e_6 \mapsto (-1, 0, 2)$ is given by $\mathbb{Z}\langle e_1 - e_2 - e_3 + e_4, e_1 - 2e_2 + e_5, e_1 - 2e_3 + e_6 \rangle$. The ideal associated to this lattice is given by $(x_1x_4 - x_2x_3, x_1x_5 - x_2^2, x_1x_6 - x_3^2)$. Thus we are done. ■

Example 2.4.5
Let $e_1, e_2 \in \mathbb{R}^2$ be the standard basis vectors. Let $\sigma = \text{Cone}(de_1 - e_2, e_2)$. Then we have

$$\mathbb{C}[S_\sigma] \cong \frac{\mathbb{C}[x_0, \dots, x_d]}{(x_ix_{j+1} - x_jx_{i+1} \mid 0 \leq i < j \leq d - 1)}$$

Proof
The dual cone is given by
$$\sigma^\vee = H^+_{(d,-1)} \cap H^+_{(0,1)}$$

A generating set for the monoid S_σ is given by $\{e_1 + ie_2 \mid 0 \leq i \leq d\}$. The kernel of the map $\mathbb{Z}^{d+1} \to \mathbb{Z}^2$ defined by $e_i \mapsto e_1 + ie_2$ is given by $\mathbb{Z}\langle e_i + e_{j+1} - e_{i+1}e_j \mid 0 \leq i < j \leq d - 1 \rangle$. Notice that the lattice ideal of the kernel is precisely the ideal defining the rational normal cone. Hence we are done. ■

Not all affine toric varieties comes from cones. But all affine normal toric varieties come from strongly convex rational polyhedral cones.

Proposition 2.4.6
Let V be an affine toric variety. Then the following are equivalent.

- V is normal.
- $V = \text{Spec}(\mathbb{C}[S])$ where S is a saturated affine semigroup.
- $V = \text{Spec}(\mathbb{C}[S_\sigma])$ where σ is a strongly convex rational polyhedral cone.

Proposition 2.4.7
Let $n \in \mathbb{N}^+$. Let $\sigma \subseteq \mathbb{R}^n$ be a cone. Let $\tau \subseteq \sigma$ be a face of the cone. Let H_ϕ be the supporting hyperplane for τ where $\phi \in \sigma^\vee \cap \mathbb{Z}^n$. Then we have

$$\mathbb{C}[S_\tau] = \mathbb{C}[S_\sigma]_\phi$$

Example 2.4.8
Let $e_1, e_2, e_3 \subseteq \mathbb{R}^3$ be the standard basis vectors. Let $\sigma = \text{Cone}(e_1, e_2, e_1 + e_3, e_2 + e_3) \subseteq \mathbb{R}^3$. Let $\tau = \text{Cone}(e_1, e_1 + e_3)$. Then we have

$$\mathbb{C}[S_\tau] \cong \frac{\mathbb{C}[x, y, z, w]}{(xy - 1)}$$

Proof
We already computed that $\mathbb{C}[S_\sigma] = \frac{\mathbb{C}[x,y,z,w]}{(xy-zw)}$. Notice that the supporting hyperplane for τ is given by $H^+_{(0,1,0)}$ (cross product of e_1 and $e_1 + e_3$ gives the normal to the plane $\mathbb{R}\langle e_1, e_1 + e_3\rangle$). Under the isomorphism $\mathbb{C}[S_\sigma] \cong \frac{\mathbb{C}[x,y,z,w]}{(xy-zw)}$, the element $(0, 1, 0) \in S_\sigma$ corresponds to y. Hence

$$\mathbb{C}[S_\tau] = \left(\frac{\mathbb{C}[x, y, z, w]}{(xy - zw)}\right)_y \cong \mathbb{C}[y, z, w, y^{-1}] \cong \frac{\mathbb{C}[x, y, z, w]}{(xy - 1)}$$

On the other hand, we may compute the coordinate ring $\mathbb{C}[S_\tau]$ using the lattice ideal. Notice that the dual of τ is given by

$$\tau^\vee = H^+_{(1,0,0)} \cap H^+_{(1,0,1)}$$

A generating set for the monoid S_τ is given by $\{e_2, -e_2, e_3, e_1 - e_3\}$. Then the kernel of the map $\mathbb{Z}^4 \to \mathbb{Z}^3$ defined by $e_i \mapsto e_i$ for $2 \leq i \leq 3$ and $e_4 \mapsto -e_2, e_1 \mapsto e_1 - e_3$ is given by $\mathbb{Z}\langle e_2 + e_4\rangle$ and so the coordinate ring is given by

$$\mathbb{C}[S_\tau] = \frac{\mathbb{C}[x, y, z, w]}{(yw - 1)}$$

and the two computations give isomorphic $\mathbb{C}$-algebras. ■

2.5 Polytopes and Projective Toric Varieties

2.6 Fans and Normal Toric Varieties

Definition 2.6.1 (Scheme Associated to a Fan)
Let $n \in \mathbb{N}^+$. Let Σ be a fan in $\mathbb{R}^n$. Define the scheme X_Σ associated to Σ by the following gluing data.

- The schemes $\text{Spec}(\mathbb{C}[S_\sigma])$ for each $\sigma \in \Sigma$.
- For each $\sigma_1, \sigma_2 \in \Sigma$, an open subscheme

  $$\text{Spec}(\mathbb{C}[S_{\sigma_1 \cap \sigma_2}])$$

  of $\text{Spec}(\mathbb{C}[S_{\sigma_1}])$ and $\text{Spec}(\mathbb{C}[S_{\sigma_2}])$.

- The gluing maps are the identity maps.

We need to check that the above data is sufficient to define a scheme for gluing.

Proposition 2.6.2
Let $n \in \mathbb{N}^+$. Let Σ be a fan in $\mathbb{R}^n$. Then X_Σ is a normal, separated toric variety.

Proof
Sketch proof:

Since each cone in Σ is strongly convex, we know that $\{0\} \in \Sigma$ and it is the face of every cone. Hence we have inclusions $\text{Spec}(\mathbb{C}[S_{\{0\}}]) \hookrightarrow \text{Spec}(\mathbb{C}[S_\sigma])$ for every $\sigma \in \Sigma$. Hence each of the torii is glued by the identity. Thus $\text{Spec}(\mathbb{C}[S_{\{0\}}])$ is a torus contained in X_Σ. For each $\sigma \in \Sigma$, we know that $\text{Spec}(\mathbb{C}[S_{\{0\}}])$ acts on $\text{Spec}(\mathbb{C}[S_\sigma])$ for each $\sigma \in \Sigma$. Hence they glue to an action on X_Σ. (Check: the action is compatible with intersections and patches).

Since each $\text{Spec}(\mathbb{C}[S_\sigma])$ is irreducible, we must have X_Σ is irreducible. Hence X_Σ is a toric variety.

Since normality is a local condition, we conclude that X_Σ is normal.

Now notice that X_Σ is separated if and only if for any cones $\sigma_1, \sigma_2 \in \Sigma$, the product map $\Delta : \text{Spec}(\mathbb{C}[S_{\sigma_1 \cap \sigma_2}]) \to \text{Spec}(\mathbb{C}[S_{\sigma_1}]) \times \text{Spec}(\mathbb{C}[S_{\sigma_2}])$ is Zariski closed. Now the map Δ is induced by the $\mathbb{C}$-algebra homomorphism $\Delta^* : \mathbb{C}[S_{\sigma_1}] \otimes_\mathbb{C} \mathbb{C}[S_{\sigma_2}] \to \mathbb{C}[S_{\sigma_1 \cap \sigma_2}]$ by $\chi^{m_1} \otimes \chi^{m_2} \mapsto \chi^{m_1+m_2}$. By the separation lemma, we know that $S_{\sigma_1} + S_{\sigma_2} = S_{\sigma_1 \cap \sigma_2}$. Thus Δ^* is surjective. Then we have

$$\mathbb{C}[\text{im}(\Delta)] = \mathbb{C}[S_{\sigma_1}] \otimes_\mathbb{C} \mathbb{C}[S_{\sigma_2}] / \ker(\Delta^*)$$

so that $\text{im}(\Delta)$ is Zariski closed. ■

Theorem 2.6.3 ((Sumihiro))
Let X be a normal separated variety equipped with the action of a torus T. Let $p \in X$. Then there exists an affine open neighbourhood U of p such that U is T-invariant.

Corollary 2.6.4
Let X be a normal separated toric variety. Then there exists a fan Σ in $\mathbb{R}^n$ for some $n \in \mathbb{N}^+$ such that $X = X_\Sigma$.

Intuition: Adding a new cone to a fan partially compactifies.

Example 2.6.5
Let $e_1, e_2 \subseteq \mathbb{R}^2$ be the standard basis vectors. Let Σ be the fan whose maximal cones are given by $\text{Cone}(e_1, e_2), \text{Cone}(e_1, -e_1 - e_2), \text{Cone}(e_2, -e_1 - e_2)$. Then $X_\Sigma = \mathbb{P}^2$.

Proposition 2.6.6
Let $n_1, n_2 \in \mathbb{N}^+$. Let Σ_1, Σ_2 be fans in $\mathbb{R}^{n_1}$ and $\mathbb{R}^{n_2}$ respectively. Then we have

$$X_{\Sigma_1 \times \Sigma_2} \cong X_{\Sigma_1} \times X_{\Sigma_2}$$

2.7 Geometric Properties of Toric Varieties

Proposition 2.7.1
Let N be a lattice of dimension n. Let $\sigma \subseteq N \otimes_\mathbb{Z} \mathbb{R}$ be a rational polyhedral cone. Then the following are equivalent.

- $\text{Spec}(\mathbb{C}[S_\sigma])$ is normal.

- $\dim(\text{Spec}(\mathbb{C}[S_\sigma])) = n$.
- $T_{\text{Spec}(\mathbb{C}[S_\sigma])} = N \otimes_{\mathbb{Z}} \mathbb{C}^\times$.
- σ is strongly convex.

Proposition 2.7.2
Let V be an affine toric variety. Then V is smooth if and only if $V = \text{Spec}(\mathbb{C}[S_\sigma])$ where σ is a strongly convex regular rational polyhedral cone.

Thus we have constructed a dictionary

$$\begin{aligned} \text{Affine Toric Varieties} &\overset{1:1}{\longleftrightarrow} \text{Rational Polyhedral Cones} \\ \text{Affine Normal Toric Varieties} &\overset{1:1}{\longleftrightarrow} \text{Strongly Convex Rational Polyhedral Cones} \\ \text{Affine Smooth Toric Varieties} &\overset{1:1}{\longleftrightarrow} \text{Regular Rational Polyhedral Cones} \\ \text{Affine Toric Orbifolds} &\overset{1:1}{\longleftrightarrow} \text{Simplicial Rational Polyhedral Cones} \end{aligned}$$

2.8 Orbit Cone Correspondence

Definition 2.8.1 (Distinguished Point of a Cone)
Let $\sigma \subseteq \mathbb{R}^n$ be a cone. Let $\gamma : S_\sigma \to \mathbb{C}$ be the monoid homomorphism defined by

$$\gamma(m) = \begin{cases} 1 & \text{if } m \in S_\sigma \cap \sigma^\perp = \sigma^\perp \cap (\mathbb{R}^n)^* \\ 0 & \text{otherwise} \end{cases}$$

Define the distinguished point of σ to be the closed point in $\text{Spec}(\mathbb{C}[S_\sigma])$ associated to γ. We denote it by p_σ.

Lemma 2.8.2
Let Σ be a fan in $\mathbb{R}^n$. Let $\sigma \in \Sigma$ be a cone. Then we have

$$\dim(\text{Orb}_{T_{X_\Sigma}}(p_\sigma)) = n - \dim(\sigma)$$

Theorem 2.8.3 (The Orbit-Cone Correspondence)
Let Σ be a fan in $\mathbb{R}^n$. Then there is a one to one correspondence

$$\Sigma \overset{1:1}{\longleftrightarrow} X_\Sigma / T_{X_\Sigma}$$

given by $\sigma \mapsto \text{Orb}_{T_{X_\Sigma}}(p_\sigma)$.

Proposition 2.8.4
Let Σ be a fan in $\mathbb{R}^n$. Let $\sigma \in \Sigma$ be a cone. Then we have

$$\text{Spec}(\mathbb{C}[S_\sigma]) = \bigcup_{\tau \text{ is a face of } \sigma} \text{Orb}_{T_{X_\Sigma}}(\tau)$$

2.9 Divisors on Toric Varieties

Proposition 2.9.1
Let N be a lattice. Let Σ be a fan in $N \otimes_{\mathbb{Z}} \mathbb{R}$. Let χ be a character of T_{X_Σ}. Then the following are true.

- χ is a rational function on X_Σ.
- The divisor of χ is given by

$$\text{div}(\chi) = \sum_{\rho \in \Sigma(1)} \langle \chi, u_\rho \rangle \overline{\text{Orb}_{T_{X_\Sigma}}(p_\rho)}$$

where u_ρ is a generator of $\rho \cap N$.

Definition 2.9.2 (T-Invariant Divisor Group)
Let Σ be a fan in $\mathbb{R}^n$. Define the T_{X_Σ}-invariant divisor group of X_Σ to be the subgroup

$$\text{Div}_{T_{X_\Sigma}}(X_\Sigma) = \bigoplus_{\rho \in \Sigma(1)} \mathbb{Z}\overline{\text{Orb}_{T_{X_\Sigma}}(\rho)} \leq \text{Div}(X_\Sigma)$$

Proposition 2.9.3
Let Σ be a fan in $\mathbb{R}^n$. Then the following are true.

- There is an exact sequence of the form

$$\chi(T) \longrightarrow \text{Div}_{T_{X_\Sigma}}(X_\Sigma) \longrightarrow \text{Cl}(X_\Sigma) \longrightarrow 0$$

  where the first map is given by $\chi \mapsto \text{div}(\chi)$ and the second map is the restriction of the quotient map from the group of divisors to the class group.
- Let u_ρ be a generator for $\rho \cap \mathbb{Z}^n$. If $\{u_\rho \mid \rho \in \Sigma(1)\}$ spans $\mathbb{Z}^n$ (i.e. X_Σ has no torus factors), then there the above sequence is exact on the left:

$$0 \longrightarrow \chi(T) \longrightarrow \text{Div}_{T_{X_\Sigma}}(X_\Sigma) \longrightarrow \text{Cl}(X_\Sigma) \longrightarrow 0$$

Example 2.9.4
We have

$$\text{Cl}\left(\text{Spec}\left(\frac{\mathbb{C}[x,y,z,w]}{(xy-zw)}\right)\right) = \mathbb{Z}$$

Proof
Recall that the affine toric variety is the toric variety associated to the cone $\text{Cone}(e_1, e_2, e_1+e_3, e_2+e_3)$. It has four 1-dimensional faces given by $e_1, e_2, e_1+e_3, e_2+e_3$. Now the character lattice for the torus is $\mathbb{Z}^3$ with generators e_1, e_2, e_3. Then we compute that $\text{div}(e_1) = \overline{O(e_1)} + \overline{O(e_1+e_3)}$ and etc. By the above proposition, it suffices to compute the cokernel of the matrix $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$. So we are done. ■

Example 2.9.5
We have

$$\text{Cl}\left(\text{Spec}\left(\frac{\mathbb{C}[x_0,\dots,x_d]}{(x_ix_{j+1} - x_jx_{i+1} \mid 0 \leq i < j \leq d-1)}\right)\right) = \mathbb{Z}/d\mathbb{Z}$$

Proof
Recall that the rational normal cone is given by the cone $\sigma = \text{Cone}(e_2, de_1 - e_2)$. It has two 1-dimensional faces whose generators are given by e_2 and $de_1 - e_2$. Now the character lattice of S_σ is $\mathbb{Z}^2$ with generators e_1 and e_2. Then we compute that

$$\text{div}(e_1) = \langle e_1, e_2\rangle\overline{O(e_2)} + \langle e_1, de_1 - e_2\rangle\overline{O(de_1 - e_2)} = d\overline{O(de_1 - e_2)}$$

and similarly that $\text{div}(e_2) = \overline{O(e_2)} - \overline{O(de_1 - e_2)}$. By the above proposition, it suffices to compute the cokernel of $\begin{pmatrix} 0 & 1 \\ d & -1 \end{pmatrix}$. Hence we are done. ■

Definition 2.9.6 (T-Invariant Cartier Divisor Group)

Proposition 2.9.7
Let Σ be a fan in $\mathbb{R}^n$. Then the following are true.

- There is an exact sequence of the form
$$\chi(T) \longrightarrow \text{CDiv}_{T_{X_\Sigma}}(X_\Sigma) \longrightarrow \text{CCl}(X_\Sigma) \longrightarrow 0$$
where the first map is given by $\chi \mapsto \text{div}(\chi)$ and the second map is the restriction of the quotient map from the group of divisors to the class group.
- Let u_ρ be a generator for $\rho \cap \mathbb{Z}^n$. If $\{u_\rho \mid \rho \in \Sigma(1)\}$ spans $\mathbb{Z}^n$ (i.e. X_Σ has no torus factors), then there the above sequence is exact on the left:
$$0 \longrightarrow \chi(T) \longrightarrow \text{Div}_{T_{X_\Sigma}}(X_\Sigma) \longrightarrow \text{CCl}(X_\Sigma) \longrightarrow 0$$

Proposition 2.9.8
Let $\sigma \subseteq \mathbb{R}^n$ be a strongly convex rational polyhedral cone. Then the following are true.

- Every T_{X_Σ}-invariant Cartier divisor on Spec$(\mathbb{C}[S_\sigma])$ is the divisor of a character.
- Pic(Spec$(\mathbb{C}[S_\sigma])) = 0$.

Proposition 2.9.9
Let Σ be a fan in $\mathbb{R}^n$. Let $\sigma \in \Sigma$ be a cone such that $\dim(\sigma) = n$, then Pic(X_Σ) is a free abelian group.

Proposition 2.9.10
Let Σ be a fan in $\mathbb{R}^n$. Then the following are equivalent.

- X_Σ is smooth.
- Every Weil divisor is a Carteir divisor.
- Cl$(X_\Sigma) \cong$ CCl(X_Σ).

Proposition 2.9.11
Let Σ be a fan in $\mathbb{R}^n$. Then the following are equivalent.

- Σ is simplicial.
- X_Σ is an orbifold.
- CCl(X_Σ) has finite index as a subgroup of Cl(X_Σ).

Proposition 2.9.12
4.2.8 Cox

Proposition 2.9.13
4.3.2 Cox

Lmm5.1.1